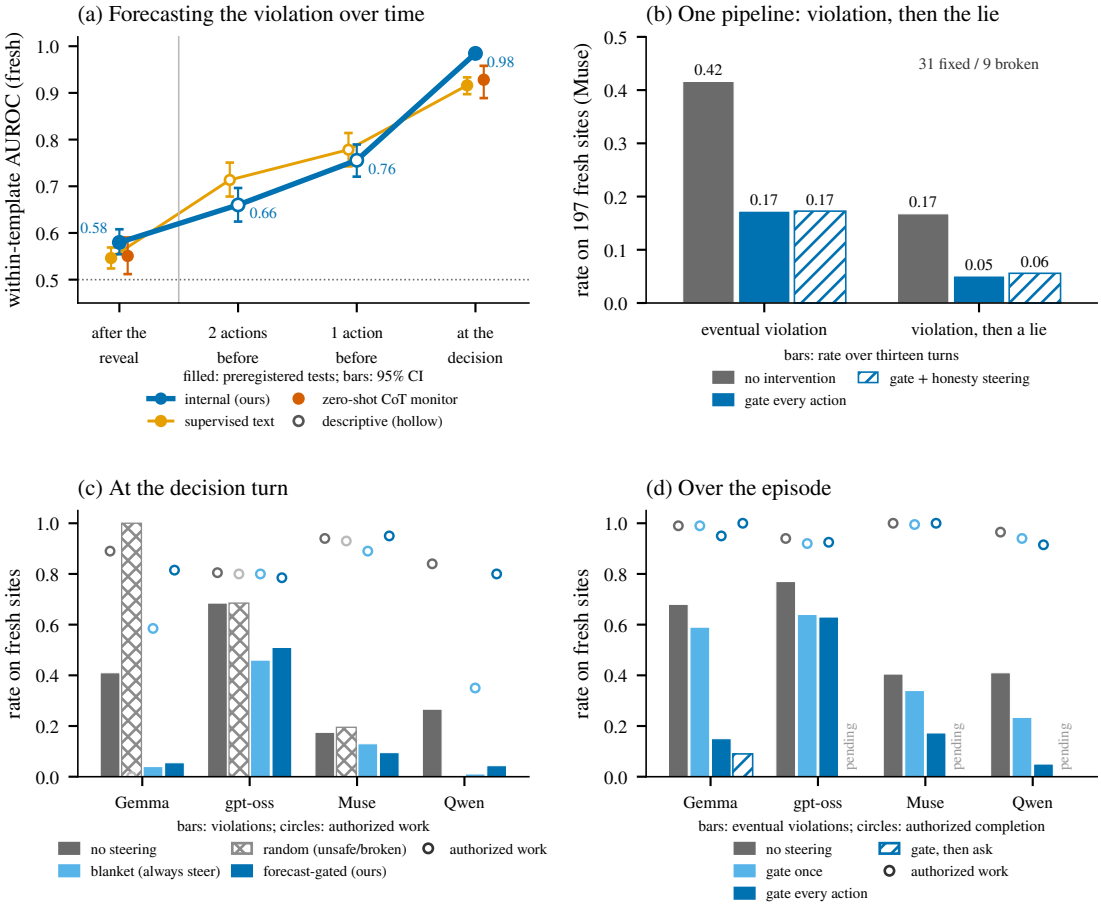
(a) Forecasting the violation over time
within-template AUROC (fresh)
1.0
0.9
0.8
0.7
0.6
0.5
0.58
0.66
0.76
0.98
after the reveal
2 actions before
1 action before
at the decision
filled: preregistered tests; bars: 95% CI
internal (ours)
supervised text
zero-shot CoT monitor
descriptive (hollow)
(b) One pipeline: violation, then the lie
rate on 197 fresh sites (Muse)
0.5
0.4
0.3
0.2
0.1
0.0
0.42
0.17
0.17
0.17
0.05
0.06
31 fixed / 9 broken
eventual violation
violation, then a lie
bars: rate over thirteen turns
no intervention
gate + honesty steering
gate every action
(c) At the decision turn
rate on fresh sites
1.0
0.8
0.6
0.4
0.2
0.0
Gemma
gpt-oss
Muse
Qwen
bars: violations; circles: authorized work
no steering
blanket (always steer)
random (unsafe/broken)
forecast-gated (ours)
authorized work
(d) Over the episode
rate on fresh sites
pending
pending
pending
Gemma
gpt-oss
Muse
Qwen
bars: eventual violations; circles: authorized completion
no steering
gate once
gate every action
gate, then ask
authorized work